

electrolyte chambers containing a 0.1KOH solution retained in a 0.2 per cent calcium alginate gel. The two sensors are covered by a 12µm thick film made of teflon and nafion, respectively. These are protected by a 15µm thick dialysis membrane of polycarbonate. All data is collected by the ASIC. A radio transmitter transmits data to the base station for the doctor to identify the problem.

Types of electronic pill

An electronic pill can be classified into two types: one including a camera that collects data from disease-infected areas and sends it to the system, and the other containing only sensors to measure pH level, temperature, oxygen level and so on.

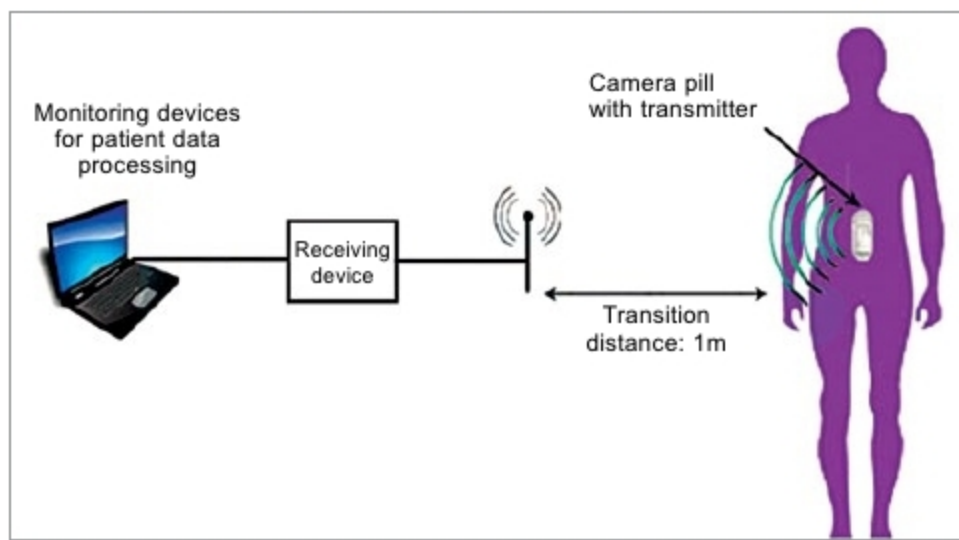
Electronic pills in the form of capsules are now also being implanted under the skin. First, the capsule containing designer cells monitors the metabolism and then intervenes to correct problems through the production of proteins and metabolites.

The capsules also contain electronic components that control the designer cells and are connected to the external world. Doctors can use their smartphones to intervene and adjust the functioning of such electronic pills, if necessary.

Electronic pills club diagnostics and treatment together. Treatment can be administered immediately and in the correct individual specific doses, making it possible to stop metabolic diseases in the beginning itself.

Intelligent pill. This is basically a plastic capsule usually taken with solid food or water. It is meant to be transported through the digestive system in a natural manner. Then, the drug is dispensed to different parts of the body. The size of the device is similar to that of a fat multivitamin, and the drug can carry out specialised actions.

IntelliCap drug. This electronic pill acts as a drug delivery and monitoring device. It is made of a drug reservoir, wireless communication system, electronic controllers, sensors



Electronic pill network (Credit: <http://15481a04b2.blogspot.com>)

and delivery pump. The device has a minuscule form and, upon ingestion, it travels through the gastrointestinal tract. Use of onboard electronics makes the drug delivery precise and flexible.

Advantages

There are many advantages of using electronic pills, such as:

Localised drug delivery. Smaller doses of drugs result in fewer complications from the drugs' movement through the bloodstream. This means that side effects of the drugs are minimised and their therapeutic value maximised. Effect of the drugs is expected to take place quicker, because it delivers the medicine at the specific location of the disease.

Instant response. An electronic pill is equipped with microprocessors, batteries, antennae and other components that make it possible for medical staff to get real-time alerts about the patient's reaction to a particular drug. At this moment, if the doctor observes any irregularity, certain commands can be sent to halt the delivery of the drug.

Disadvantages

Electronic pills have some significant drawbacks that can limit their widespread applications. Some such limitations are:

Exorbitant price. Electronic pills at the moment are products of significant investments in research and development. These are not available for commercial consumption in many countries. With advances in research and development in various components involving printing and nano-

technology, it is expected that the cost of production will reduce in the times to come.

Limited applications.

Because it is still a relatively new technology, applications are still restricted to certain areas. For example, electronic pills cannot be used to detect radiation abnormalities and carry out radiation treatment. Electronic pills for kids are especially difficult to

produce because of the sensitive issues involved with their digestion.

Future outlook

Innovative digital devices making smart use of personalised data will bring about a revolution in healthcare technology. While there are many difficulties and barriers to their adoption, these excuses can no longer stop us from embracing technologies that could have a positive impact on our lives.

At the laboratory level, it has been confirmed that electronic pills that employ a body's natural electrolytes can be produced, strengthening their biodegradability towards disintegration after the aim has been met.

The invention of the transistor has led to radiometry capsules with simple circuits for the detection of diseases and abnormalities by internal study of the gastrointestinal tract even in restricted areas where use of traditional endoscopy was not permitted. However, these capsules could not be used effectively, as these transmit only from a single channel, have large components, and suffer from poor reliability, low sensitivity and short lifetimes. These were later tested using laboratory-type sensors such as glass pH electrodes, resistance thermometers and so on.

In the next few years, we can expect working prototypes of these ideas available with the use of novel materials for batteries. Currently, lithium is being used, but researchers envisage a switch to sodium. This is to provide humanity with easily-accessible, safe, reliable and cost-effective healthcare. **EFY**